

Skills Summary

Quadrilaterals: Which Way Does the Implication Go?

Use this reference while completing your worksheet or when studying.

The family tree runs one way. square $\Rightarrow$ rhombus $\Rightarrow$ parallelogram $\Rightarrow$ quadrilateral, and square $\Rightarrow$ rectangle $\Rightarrow$ parallelogram. Reading *down* the tree is always safe; reading *up* it needs an extra property, and that extra property is what every problem on the worksheet is testing.

1. What a property is enough for

Upgrades that work (each is a genuine theorem):

parallelogram + one right angle $\Rightarrow$ rectangle parallelogram + congruent diagonals $\Rightarrow$ rectangle

parallelogram + two congruent adjacent sides $\Rightarrow$ rhombus parallelogram + perpendicular diagonals $\Rightarrow$ rhombus

rhombus + one right angle $\Rightarrow$ square rectangle + congruent adjacent sides $\Rightarrow$ square

Not enough: congruent sides alone never gives right angles, and congruent diagonals alone never gives congruent sides.

Example: A rhombus has consecutive angles $(3x + 20)^\circ$ and $(2x + 10)^\circ$. Find x , then decide whether it must be a square.

Step 1. A rhombus is a parallelogram, so consecutive angles are supplementary — that is the equation to write, and the resulting angles are what decide “square”.

Step 2. $(3x + 20) + (2x + 10) = 180$, so $5x + 30 = 180$. $\Rightarrow$ $x = 30$, giving angles 110° and 70° : not right angles, so a rhombus and *not* a square.

Try it: A rhombus has consecutive angles $(4x)^\circ$ and $(5x)^\circ$. Find x .

check: $9x = 180$

2. Testing a claim with side lengths

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Two *adjacent* sides of unequal length kill “rhombus” and “square” instantly — one counterexample side pair is all it takes. Perpendicular diagonals give a rhombus only when they also bisect each other (equal midpoints); without that you have a kite.

Example: $A(0, 5)$, $B(2, 0)$, $C(0, -3)$, $D(-2, 0)$: the diagonals lie on the axes, so they are perpendicular. Must $ABCD$ be a rhombus?

Step 1. Perpendicular diagonals are not sufficient on their own, so test two adjacent sides rather than trusting the claim.

Step 2. $AB = \sqrt{2^2 + 5^2} = \sqrt{29}$, while $BC = \sqrt{2^2 + 3^2} = \sqrt{13} \approx 3.61$. $\Rightarrow$ $AB \approx 5.39 \neq BC$, so not a rhombus — it is a kite.

Try it: $A(0, 7)$, $B(2, 0)$, $C(0, -4)$, $D(-2, 0)$. Find AB to two decimal places (and see that BC differs).

check: 8.27 becomes 68.9

3. The diagonal test for the most specific name

Rule: once all four sides are known congruent (a rhombus), compare the *diagonals*:
 diagonals congruent $\Rightarrow$ square diagonals not congruent $\Rightarrow$ rhombus only.

The same comparison run on a rectangle tells you nothing new — a rectangle always has congruent diagonals. Order matters: sides first, then diagonals.

Example: $A(0, 0)$, $B(3, 1)$, $C(4, -2)$, $D(1, -3)$ has four congruent sides. Square or rhombus?

Step 1. All sides congruent already gives a rhombus, so the deciding test is whether the diagonals are congruent.

Step 2. $AC = \sqrt{4^2 + 2^2} = \sqrt{20}$ and $BD = \sqrt{2^2 + 4^2} = \sqrt{20}$ — equal. $\Rightarrow AC = 4.47 = BD$, so $ABCD$ is a square.

Try it: The rhombus $E(0, 0)$, $F(4, 2)$, $G(8, 0)$, $H(4, -2)$. Find diagonal EG , compare it with $FH = 4$, and name the figure.

check: $EG = 8$, so rhombus only

4. Killing a converse with a counterexample

Rule: to show an implication does *not* run backwards, produce one figure with the property but without the name. The classic pairs:

perpendicular diagonals without a rhombus $\rightarrow$ **kite** congruent diagonals without a rectangle $\rightarrow$ **isosceles trapezoid**

The missing hypothesis in both is “and it is a parallelogram”.

Example: Is “congruent diagonals $\Rightarrow$ rectangle” always true? Test $T(0, 0)$, $U(6, 0)$, $V(5, 3)$, $W(1, 3)$.

Step 1. To refute a converse I only need one figure with congruent diagonals that is not a rectangle, so compute both diagonals of this trapezoid.

Step 2. $TV = \sqrt{5^2 + 3^2} = \sqrt{34}$ and $UW = \sqrt{5^2 + 3^2} = \sqrt{34}$: congruent, yet TU and WV are the only parallel pair and no angle is right. $\Rightarrow TV = 5.83 = UW$ in a shape that is *not* a rectangle.

Try it: $T(0, 0)$, $U(8, 0)$, $V(6, 4)$, $W(2, 4)$. Find TV (then check UW matches).

check: 17.2